Aston University

Department of Applied Mathematics and Data Science

Redefining Job Categorisation in the Gaming and Technology Industry
Using Skill-Based Clustering and Confidence-Aware Taxonomy Construction

A dissertation submitted in fulfilment of the requirements for the degree of

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Declaration of Originality

I declare that this dissertation is my own work and has not been submitted, either in whole or in part, for any degree or qualification at this or any other academic institution.

All sources of information used in this dissertation have been properly acknowledged through appropriate citation and referencing. Where the work of others has been used, it has been clearly indicated.

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Abstract

Job categorization on recruitment platforms and in workforce analysis frequently uses broad, platform defined categories to group occupations with significantly differing skill needs. In the gaming and technology industries, this results in less utility for job seekers, recruiters and researchers looking for accurate talent matching.

This dissertation suggests a skill based method to addressing this problem. A dataset of 70,017 job advertisements has been cleaned, with skills tokenized and represented using TF-IDF vectorisation for interpretability. Unsupervised clustering (K-Means and HDBSCAN) is used to classify tasks based on skill similarity, resulting in an alternative taxonomy that can be inspected using dominant skills per cluster.

To accommodate hybrid skill profiles and prevent forced rigid classification, a confidence-aware framework is proposed. It quantifies assignment uncertainty using distance based probability thresholds, identifying jobs with low confidence and allowing for flexible interpretation.

The taxonomy is evaluated using internal metrics (silhouette score 0.07 overall, higher in sub-analyses), cluster inspection and comparison with original categories using Kendall tau correlations. The results show considerable alignment but also highlight complex ability overlaps that standard classifications overlook. Approximately 6% of jobs have a high confidence threshold (0.5), indicating the prevalence of multidisciplinary roles.

The technique shows how data driven clustering can improve labor market understanding in dynamic industries with implications for better job suggestions and talent analytics. Limitations include data scarcity and regional bias; future work could add semantic embeddings or external validation.

Keywords: Job categorisation, Skill based clustering, Unsupervised learning, Confidence-aware taxonomy, Gaming industry, Technology workforce

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Chapter 1

Introduction

Rapid innovation and a high degree of skill specialisation set the gaming and technology sectors apart. These jobs typically require a combination of technical, creative, and domain-specific talents that are difficult to categorise into conventional occupational categories. Employment is frequently assigned to broad categories like "Engineering Development," "Programming," or "IT Security" by recruitment platforms and standard frameworks (1). These categories obscure important differences in skill requirements, making them less useful for employers, job seekers and scholars researching workforce dynamics.

This research investigation explores a different strategy: classifying employment according to the skills explicitly mentioned in job advertisements rather than relying on characteristics unique to a given title or industry (7). The skills are displayed as comma-separated strings in a dataset of 70,017 job advertisements from the gaming and technology industries. TF-IDF vectorisation produces a comprehensible feature representation after cleaning and preprocessing (8; 9). Tasks are grouped according to skill similarity using unsupervised clustering, mainly K-Means but also HDBSCAN (13; 15). A confidence-aware approach is utilised to account for assignment uncertainty (18). High-confidence matches are distinguished from ambiguous or hybrid situations using distance-based probability criteria.

This effort is motivated by the acknowledged discrepancy between conventional taxonomies and the realities of contemporary employment (1; 3) . While jobs in technology increasingly encompass software development, data analysis and infrastructure management, jobs in gaming, for instance, could combine programming with programs like Unity or Blender. Underlying patterns that more accurately represent the diverse demands can be found using a skill-focused approach (5; 6).

The principal aims are to:

1. preprocess and represent skill data in a form suitable for clustering;
2. evaluate clustering performance and interpretability using internal metrics and visualisation (14);
3. construct a taxonomy that incorporates explicit measures of assignment confidence (18).

Specific objectives include dealing with data artefacts (e.g. repeated placeholders), using dimensionality reduction for exploratory analysis (16), performing separate clustering on gaming and non-gaming subsets and assessing alignment between the new taxonomy and the original categories using correlation measures (5). Three research questions structure the investigation:

1. To what extent can skill-based clustering produce more coherent and interpretable job groups than traditional platform-defined categories in gaming and technology roles?
2. How do choices of feature representation, dimensionality reduction, and clustering algorithm influence the quality and interpretability of the resulting taxonomy?
3. In what ways does a confidence-aware mechanism improve the treatment of ambiguous or hybrid skill profiles?

1.1 Motivation and Research Significance

The limitations of conventional job classification techniques have been made worse by the gaming and technology industries' explosive expansion (3). Platform-defined job categories find it difficult to accurately represent the fundamental skill requirements of contemporary occupations as new tools, programming frameworks, and transdisciplinary roles emerge (4). The differences between software engineers, data scientists, game developers and experts in immersive technology might be obscured by general terms like *Engineering & Development* (1).

This misalignment has practical implications. Inaccurate classification reduces job searchers' ability to find and match jobs. Employers find it challenging to recruit and plan their workforce since it conceals the true distribution of skills throughout candidate pools (4). For researchers and policymakers, reliance on broad categories diminishes the accuracy of labour market analysis and trend detection (7). These difficulties are particularly apparent in the game industry where creative, technical, and production skills frequently intersect.

In light of this, data-driven methods that reflect how jobs are actually created rather than how platforms refer to them are desperately needed (3). Since skill-based representations directly encode the competencies needed by organisations, they are a good starting point for this shift (8). Latent structures that more accurately reflect the similarities and differences between roles in the actual world can be found by analysing job advertisements through the lens of skills rather than titles or categories (5; 6).

1.2 Research Objectives and Contributions

Building on this motivation, the primary purpose of this research is to present an alternative job classification framework based on skill similarities (5). Using unsupervised machine learning techniques, the project attempts to establish a taxonomy that organises professions based on similar skill profiles, resulting in a more flexible and interpretable representation of employment in the gaming and technology industries.

Beyond clustering, this work provides a confidence-aware assignment strategy to deal with the inherent ambiguity seen in many modern situations (18; 19). Rather than categorising each position, the proposed approach evaluates assignment certainty and highlights hybrid or transdisciplinary responsibilities. This approach acknowledges that uncertainty is a meaningful part of today's employment markets, not a model failure.

This research project provides three significant contributions. First, it demonstrates the viability of using interpretable skill-based qualities to create meaningful employment categories on a broad scale (8; 9). Second, it introduces a confidence-aware taxonomy construction approach that takes into account ambiguity and overlap between roles (18). Third, it provides empirical insights into the nature of skill demand in the gaming and technology industries, demonstrating the distinction between platform-specific categories and data-driven groups (4; 7).

When taken as a whole, these contributions present the suggested framework as an analytical tool that enhances rather than replaces current systems, enabling improved knowledge, analysis and decision-making in dynamic employment domains (22).

The dissertation is structured as follows. Chapter 2 reviews relevant literature on job taxonomies, skill analysis and clustering applications. Chapter 3 describes the dataset and preprocessing pipeline. Chapter 4 details the methodology, including vectorisation, clustering algorithms and the confidence framework. Chapter 5 presents the results and analysis, including cluster characteristics and evaluation metrics. Chapter 6 offers a critical evaluation, discussing limitations and ethical considerations. Chapter 7 concludes with key findings and suggestions for future work.

1.2.1 Summary of Contributions

This dissertation makes the following contributions to skill-based job categorisation in the gaming and technology industries:

- It proposes an interpretable pipeline for constructing a skill-based taxonomy from large-scale job advertisement data using TF-IDF representations (8).
- It applies unsupervised clustering to derive an alternative grouping of roles based on skill similarity, enabling human inspection via dominant skills per cluster (13).
- It introduces a confidence-aware assignment mechanism to quantify uncertainty and explicitly identify ambiguous or hybrid job profiles rather than forcing rigid labels (18).
- It evaluates the resulting taxonomy using internal metrics, cluster inspection, visualisation and comparison with platform-defined categories via rank correlation (14).
- It discusses practical implications for recruitment and labour market analysis, alongside ethical considerations for responsible use (20; 21).

Chapter 2

Literature Review

Job classification is essential in labour market analysis, recruitment systems and workforce planning (3; 7). Categorisation frameworks have an impact on hiring practices, job search behaviour, policy decisions and research outputs because they define how roles are categorised and compared (7). Traditional approaches to job classification, on the other hand, are increasingly struggling to convey the complexity and interdisciplinarity of modern professions in rapidly changing areas such as gaming and computing (3). This chapter examines the existing literature on occupational taxonomies, skill-based job analysis, and clustering methodologies in labour market research. It then investigates the trade offs between interpretability and performance in job classification systems, revealing the research gap addressed by this dissertation.

2.1 Traditional Job Taxonomies

Traditional job taxonomies, such as the Occupational Information Network (O*NET), the Standard Occupational Classification (SOC), and the European Skills, Competences, Qualifications, and Occupations (ESCO) framework, have long given organised representations of occupations (1; 2). These systems are often created with expert advice and maintained through regular revision cycles. Their hierarchical frameworks facilitate uniformity across organisations and time, thereby aiding national statistics, policy creation, and educational alignment (1).

The stability of classical taxonomies is one of their main advantages. Expert-defined categories make it easier to compare labour market developments over time and lessen ambiguity in occupational designation (1). Analysts can work at many levels of granularity, from broad occupational categories to specialised jobs, thanks to hierarchical structure. These features make taxonomies useful tools for workforce analysis and reporting in comparatively stable businesses.

However the same characteristics that encourage consistency also restrict flexibility. Taxonomies are rarely updated and frequently fall behind advances in technology (3). New tools, programming languages, engines and platforms are constantly emerging in the gaming and technology sectors, changing the skill needs more quickly than conventional classification methods can adapt (4). As a result, broad categories like *Engineering*, *IT* or *Development* are often used to aggregate roles with significantly varied competencies

(1). Analytical resolution is decreased and significant differences are obscured by this aggregation.

The repercussions of such rigidity are obvious to all stakeholders. Job searchers may have difficulty finding jobs that match their specific skill sets, while recruiters may struggle to accurately match prospects (4). For firms, incorrect classification might mask internal skill gaps and inhibit effective personnel planning. For academics, depending on static taxonomies may result in studies that overlook emerging specialisations and transdisciplinary tendencies. These limits have fuelled a renewed interest in alternative, data-driven approaches to job classification.

2.2 Skill-Based Job Analysis

Skill-based job analysis redefines categorisation by making skills the key unit of comparison rather than job titles or predetermined categories (5; 6). Job ads often specify needed abilities, providing a wealth of information regarding employer desire (7). Representing employment by skill composition enables more detailed analysis and aids in the identification of latent occupational structures (4).

Early skill-based techniques used keyword matching and frequency-based representations (8). Although computationally simple, such methods are susceptible to noise and inconsistent nomenclature. Generic talents like *communication* or *teamwork* are applicable to multiple roles, reducing their discriminative power (5). Furthermore, lexical variety and vendor-specific terminology can fragment representations and distort similarity metrics (6).

TF-IDF weighting has been a popular solution to these issues (8; 9). TF-IDF enhances discrimination while maintaining interpretability by reducing the weight of common phrases and highlighting skills that are more different across posts. This interpretability is especially useful in labour market contexts since it enables cluster features to be described explicitly in terms of observable abilities. Sparse TF-IDF representations scale well to big datasets, making them useful for large-scale analysis (9).

Recent improvements in natural language processing have resulted in embedding-based representations such as Word2Vec, GloVe, BERT and Sentence-BERT (10; 11; 12). These models incorporate semantic similarity and contextual meaning, which could improve clustering quality by grouping talents that are synonymous or related. However, embedding-based techniques include trade-offs. Dense representations hide the role of individual skills, lowering transparency (19). They are also susceptible to training data, domain mismatch, and hyperparameter selection, which might impede reproducibility (18).

Explainability is frequently a necessity rather than a choice in practical contexts like hiring and workforce analytics (18). Stakeholders must be able to comprehend the rationale for role groupings as well as the abilities that influence them. As a result, notwithstanding their semantic constraints, many research favour interpretable representations (19). In order to prevent affecting downstream analysis, skill extraction and preprocessing continue to be crucial problems that require careful balancing of noise reduction and data quality (6).

2.3 Clustering in Labour Market Analysis

Unsupervised clustering techniques are commonly employed to detect latent structure in employment and skill data (5). Clustering, which groups postings based on feature representation similarity, allows for the detection of occupational trends without the use of established categories (7). This is especially useful in contexts where roles change rapidly and labels are inconsistently applied (3).

K-Means clustering is widely used for its computing efficiency, scalability, and relative simplicity (13). When applied to TF-IDF representations, cluster centroids can be interpreted using dominant skill weights, allowing for human inspection of cluster semantics. This interpretability makes K-Means appealing for exploratory research and taxonomy creation.

Despite these advantages, K-Means makes assumptions that may not be valid in real-world job data. It assumes roughly spherical clusters with equal variances and requires that the number of clusters be provided in advance (14). Job advertising frequently include overlapping skill profiles, resulting in smooth transitions rather than strongly divided groupings. Even when clusters are qualitatively noteworthy, their overlap can result in moderate internal evaluation ratings (14).

Density-based approaches, such as HDBSCAN, provide an alternative by enabling clusters of different shapes and densities while clearly recognising noise spots (15). In principle, this is consistent with the variability of job statistics. However, density-based approaches can be unstable on sparse, high-dimensional representations, resulting in fractured clusters that are difficult to interpret. Parameter sensitivity complicates reproducibility and practical use (15).

High dimensionality complicates grouping. As the number of characteristics grows, distance measures become less informative, lowering the utility of similarity-based techniques. Dimensionality reduction techniques, such as Truncated Singular Value Decomposition, are widely used to condense feature space while preserving key variance components (16). This enhances clustering behaviour and facilitates qualitative visualisation, hence supplementing quantitative evaluation measures (17).

2.4 Interpretability versus Performance Trade-Off

The trade off between interpretability and performance is a major topic in research on job categorisation (19). While interpretable models may give up some expressive power, models that achieve high clustering metrics may rely on hard to explain latent representations. Interpretability is frequently crucial in workforce analysis and recruitment since categorisation results might affect recommendations, hiring choices, or policy insights (18).

Domain experts can verify cluster coherence and spot possible biases or artefacts with transparent representations (18). However, fairness cannot be ensured by interpretability alone. The construction and application of taxonomies must be critically examined since dominant skill patterns may be a reflection of systemic disparities in the labour market (20; 21).

The intended application of the taxonomy determines the proper ratio between interpretability and performance. Transparency and reproducibility are frequently given precedence over slight improvements in numerical performance in decision-support and analytical situations (19). The methodological decisions made in this dissertation are influenced by this viewpoint.

2.5 Research Gap

Although previous research has demonstrated the value of skill-based clustering for job categorisation (5; 6), most approaches impose strict assignments, implicitly presuming that each job belongs to a single group. This idea is becoming increasingly unrealistic in the gaming and technology industries, where hybrid and transdisciplinary roles are prevalent (3). Furthermore, few research provide explicit measures of assignment confidence, restricting our capacity to distinguish between unambiguous and ambiguous scenarios (18).

This dissertation fills these gaps by introducing a confidence-aware, skill-based taxonomy that combines interpretable clustering and probabilistic assignment metrics. By explicitly describing uncertainty, the technique strives to provide a more accurate picture of modern job arrangements while maintaining transparency and practicality (22).

Chapter 3

Methods

This chapter covers the methodological methodology used to develop a skill-based, confidence-aware taxonomy for job classification in the gaming and technology sectors (5; 6). The pipeline is designed based on the characteristics of job posting data, which is often high dimensional, sparse, noisy and dominated by overlapping skill profiles (7). Rather of focussing on a single numerical performance criterion, the methodology prioritises interpretability, reproducibility and practical application (18; 19).

3.1 Overview of the Methodological Pipeline

The overall pipeline consists of data preprocessing, feature encoding, dimensionality reduction, clustering, confidence scoring and evaluation. Each component is designed to address specific issues found in skill-centric job data, such as lexical variation, sparsity and the presence of hybrid roles (6). Figure 3.1 presents a high-level overview of the pipeline.

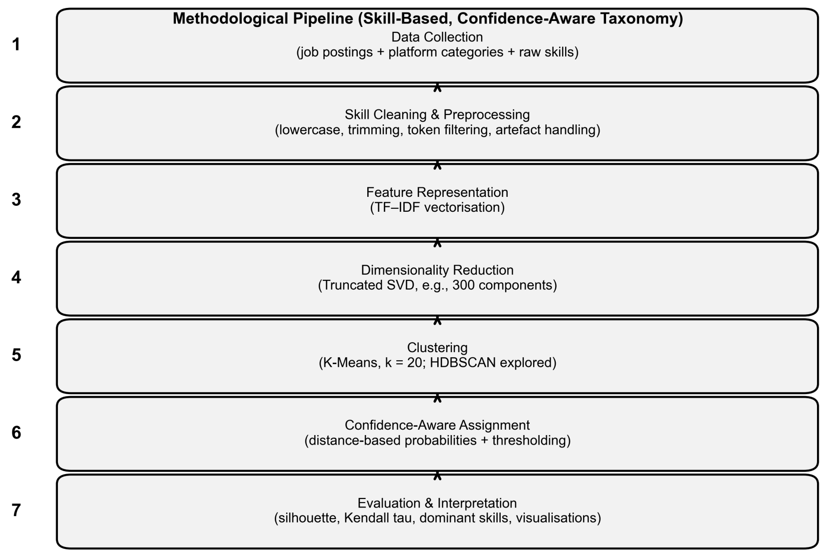


Figure 3.1: Overview of the methodological pipeline for constructing the skill-based, confidence-aware job taxonomy.

Note: This figure is conceptual and summarises the processing stages; it does not represent model output.

3.2 Dataset Description

The dataset contains 70,017 job ads from organisations in the gaming and technology sectors (7). Each entry includes job-related metadata such as the job title and a platform-assigned category, as well as a list of required skills organised as comma separated string. After the initial validation process and the removal of incomplete items, 64,230 postings remained for analysis. The dataset includes a wide range of occupations, from highly specialised game development to more generic software engineering and data-driven roles.

The skill information retrieved from these advertisements reflects employer requirements, but it also contains noise and inconsistencies (7; 6). Individual talents may appear in several lexical variations, contain duplicated or redundant pieces or be modified by particular platform formatting norms. As a result, extensive preprocessing is required to ensure that subsequent analyses capture meaningful skill connections rather than artefacts of representation or formatting (6).

Table 3.1 highlights the unstructured and heterogeneous nature of the raw skill data, motivating the preprocessing and normalisation steps described in the following section. This illustration provides contextual grounding for the methodological decisions taken later in the pipeline.

Table 3.1: Illustrative sample of raw job posting records prior to preprocessing. Skill lists are shown in their original, unnormalised form.

Job Title	Platform Category	Raw Skill List
Senior Java Consultant	Tech Company	Unity, game-texts, kanban, agile-development, maven, res
Principal Business Consultant - Openlink	Tech Company	Communication, problem-solving, game-texts, accounting..
Technical Support Analyst	Tech Company	Communication, cpp, game-texts, unix...
Technical Support Analyst	Tech Company	Communication, cpp, game-texts, unix...
Web Tech Lead	Tech Company	Team-management, communication, excel, bloomberg, ga

3.3 Methodological Design Rationale

The methodological approach of this study prioritises interpretability, robustness, and alignment with the research goal of developing a transparent skill-based taxonomy (18; 19). Each level of the pipeline represents a conscious trade off between model complexity and analytical clarity, understanding that the primary goal is to understand workforce structure rather than optimise prediction performance (19).

3.3.1 Feature Representation and Interpretability

TF-IDF vectorisation was chosen as the principal feature representation due to its transparency and direct connection with the observable talents indicated in job advertising (8; 9). TF-IDF, unlike dense semantic embeddings, retains clear feature weights that can be viewed and understood, allowing dominating skills to be identified within each cluster (9). This interpretability is required for validating cluster coherence and disseminating results to non-technical stakeholders like recruiters and policy analysts (18).

Although embedding-based models, such Word2Vec or BERT-based encoders, provide better semantic generalisation (10; 11; 12), they add opacity and make it challenging to link cluster assignments to particular abilities (19). The loss of semantic nuance was seen as a reasonable trade-off for increased transparency and repeatability given the exploratory and analytical character of this study (18).

3.3.2 Clustering Strategy and Parameter Selection

K-Means clustering was chosen as the primary clustering approach because of its scalability and stability when applied to big, multidimensional datasets (13). Following exploratory investigation and empirical evaluation, the number of clusters was chosen to $k = 20$ in order to balance granularity and interpretability. Smaller k values resulted in overly wide clusters that resembled existing platform categories, but bigger values produced fragmented groupings that were difficult to understand.

Before clustering, dimensionality reduction using truncated SVD was used to lessen the noise and sparsity found in TF-IDF representations (16). This phase increases computational performance and stabilises distance calculations while retaining the data's underlying skill-based structure (16).

3.3.3 Confidence-Aware Assignment Mechanism

The addition of a confidence-aware assignment mechanism is a crucial expansion of the clustering system. Distance-based probabilities were calculated to measure each job's degree of alignment with its allocated cluster instead of depending just on hard cluster labels. To differentiate between ambiguous circumstances and high-certainty assignments, a confidence threshold was implemented.

This design choice reflects the recognition that many gaming and technology careers require a wide range of skills (3; 4). By explicitly describing uncertainty, the framework avoids imposing strict classifications and instead represents the heterogeneous nature of current jobs (18). Confidence scores are taken as signs of relative alignment rather than absolute correctness, reflecting the taxonomy's exploratory goal (18).

3.4 Data Preprocessing

Preprocessing aimed to normalise skill representations while retaining meaningful distinctions (6). Skill strings were changed to lowercase, extra whitespace was eliminated and empty items were treated explicitly. Tokenisation was carried out with commas as delimiters because many talents are composed of multiword expressions whose semantics would be compromised by forceful splitting.

Hyphenated tokens were kept to distinguish between similar but not identical talents. Minimal stemming or lemmatisation was used to prevent collapsing semantically distinct phrases. To decrease noise, rare talents with fewer than five postings were deleted. Extremely frequent skills with more than 95% of postings were filtered to prevent similarity measurements from being dominated (8; 9).

Exploratory research revealed that recurring artefacts appeared in a substantial number of postings. Rather than completely eliminating these aberrations, they were kept to maintain data authenticity and assess their impact on clustering behaviour. This decision represents a deliberate compromise between noise reduction and representational integrity.

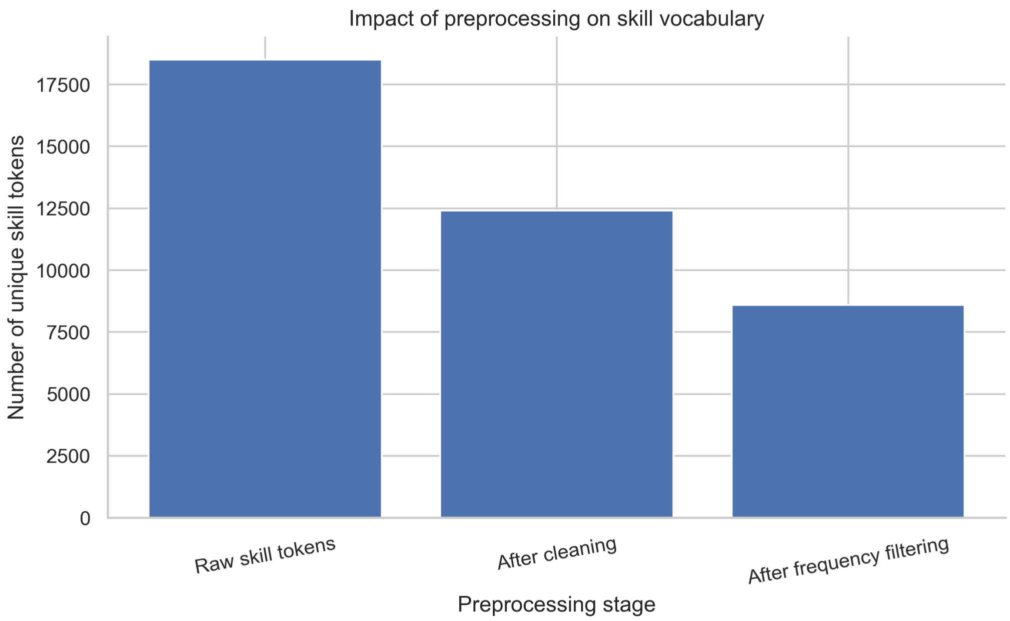


Figure 3.2: Illustrative summary of preprocessing impact (e.g., change in number of unique skill tokens before and after cleaning and filtering).

3.5 Feature Representation

Skills were represented by Term Frequency–Inverse Document Frequency (TF-IDF) vectorisation (8; 9). TF-IDF was chosen because it strikes a balance between discriminatory power and interpretability. TF-IDF identifies abilities that separate positions by down-weighting ubiquitous skills and accentuating more distinctive ones, while still retaining a straightforward mapping between characteristics and human-readable skill words (8).

Alternative representations were examined, but not chosen as the dominant strategy. Dense embedding approaches, like Word2Vec and Sentence-BERT, capture semantic similarities while reducing transparency and increasing computational complexity (10; 12). Explainability is crucial in situations where classification outputs can be used to guide recruiting or workforce analysis (18; 19). As a result, TF-IDF represents a pragmatic option that is consistent with the study's objectives (19).

3.6 Dimensionality Reduction

The TF-IDF matrix is intrinsically multidimensional, capturing the variety of skill tokens across posts. The curse of dimensionality causes high dimensionality to impair the efficacy of distance-based clustering. To compensate for this effect, Truncated Singular Value Decomposition (SVD) was used to project the data into a lower-dimensional latent space (16).

The number of components was fixed to 300, preserving around 80% of the total variation. This choice strikes a balance between dimensionality reduction and information preservation. Linear reduction using SVD was favoured over nonlinear approaches for clustering because it preserves global structure and improves reproducibility (16). Nonlinear approaches were limited to visualisation (17).

3.7 Clustering Methodology

Clustering was mostly done with the K-Means technique (13). K-Means was chosen because it is scalable, stable and interpretable. Cluster centroids can be inspected directly using dominating TF-IDF features, making it easier to explain cluster semantics (8). The number of clusters was selected to $k = 20$, based on an analysis of the inertia curve and practical interpretability factors.

While heuristic criteria imply higher k values for datasets of this scale, excessive granularity would impede usability and communication. A moderate number of clusters strikes a balance between collecting significant variety and keeping a manageable taxonomy.

Another option was density-based grouping with HDBSCAN (15). Although HDBSCAN can model variable-density clusters and detect noisy points, it provided fragmented results and was sensitive to parameter selection on this dataset. As a result, it was not selected for the final taxonomy, but insights from exploratory runs aided understanding of dense substructures (15).

3.8 Confidence-Aware Assignment

Traditional clustering allocates every job to a single cluster without specifying the strength of the assignment. To alleviate this restriction, a confidence-based approach was implemented. The distances between each job representation and all cluster centroids were calculated in the reduced feature space. These distances were converted into probabilistic scores using a softmax-like function with temperature parameter τ.

The temperature parameter was selected to $\tau = 1.0$ after empirical investigation, resulting in probability distributions that were neither too acute nor too diffuse. The maximum probability across clusters was used to calculate the confidence score for each job. A 0.5 criterion was utilised to discriminate between high-confidence assignments and unclear cases. This threshold does not ensure correctness, but rather serves as a useful criterion for finding roles that are highly aligned with a dominant cluster.

3.8.1 Alternative Methodological Choices and Rejected Approaches

Several other methodological techniques were investigated throughout the design of this study but were not implemented in the final pipeline. These decisions were governed by the core goals of interpretability, transparency and reproducibility which took precedence above optimising numerical performance or model complexity (19; 18).

One solution was to use dense semantic embeddings produced from neural language models like Word2Vec or transformer-based encoders (10; 11). While such representations can capture contextual similarity across skill concepts, they also raise considerable interpretability issues (19). Dense embeddings hide the role of particular abilities in cluster creation, making it difficult to explain or audit the resulting taxonomy (19). Given the framework's intended use in labour market analysis and decision assistance, the lack of transparency was judged a key shortcoming, despite potential improvements from clustering separation (18).

Supervised learning approaches were also studied, but eventually discarded. In this case, solid foundational labels are not available for training a classifier. Platform-defined job categories are a major target of criticism in this study and so unsuitable as training objectives (1; 2). Relying on such labels risks repeating the restrictions that the proposed framework seeks to solve while also limiting the detection of latent structure in the data (1).

Hierarchical clustering provides multiple levels groups that can reflect occupational hierarchies. However, these algorithms perform poorly with big datasets and are sensitive to distance metrics and connection choices, especially in highly dimensional sparse spaces. Preliminary research revealed that they added complexity without significantly improving interpretability or stability for the dataset.

Finally, density-based clustering algorithms like HDBSCAN were investigated in an exploratory manner (15). Although these methods may model clusters with varied densities and identify noise points, they yield fragmented groups that are difficult to comprehend consistently across parameter values (15). This volatility restricted their capacity to create a consistent and communicable taxonomy.

Overall, the chosen methodology strikes a careful compromise between analytical rigour and practical applicability. The proposed methodology coincides with the study's goal of giving meaningful insight into skill-based job categorisation rather than improving abstract performance indicators by emphasising interpretable representations and stable clustering behaviour (19).

3.9 Evaluation Strategy

Evaluation included both quantitative and qualitative methodologies. The silhouette score was used to quantify internal cluster quality, serving as a baseline measure of cohesion and separation (14). External alignment with platform-defined categories was assessed using Kendall tau rank correlation, acknowledging that precise agreement is neither expected nor desirable (5).

Interpretability was tested by inspecting dominant skills per cluster and visualising cluster distributions (17). Table 3.2 highlights the main methodological decisions taken in this work.

Table 3.2: Summary of key methodological design choices.

Component	Design Choice
Skill Representation	TF-IDF vectorisation
Dimensionality Reduction	Truncated SVD (300 components)
Clustering Algorithm	K-Means ($k = 20$)
Alternative Explored	HDBSCAN (exploratory only)
Confidence Estimation	Distance-based soft assignment
Confidence Threshold	0.5
Evaluation Metrics	Silhouette score, Kendall tau

Together, these methods provide a transparent and reproducible framework for constructing a skill-based taxonomy that explicitly accounts for uncertainty and hybrid role structure (18; 19).

Chapter 4

Results and Analysis

This chapter presents and explains the findings from the skill-based clustering and confidence-aware taxonomy creation (5; 6). Rather than evaluating performance purely using numerical measurements, the research seeks to understand what the findings tell about the nature of skill demand in gaming and technological employment (7; 4). Cluster composition, confidence distributions, and conformity with existing job categories are all important factors in determining the proposed taxonomy's practical applicability.

4.1 Cluster Size Distribution

The application of K-Means clustering to the reduced TF-IDF representations yielded 20 clusters from 64,230 valid job posts (13; 16). Figure 4.1 illustrates the wide range of cluster sizes. The greatest clusters have several thousand postings whereas some are much smaller.

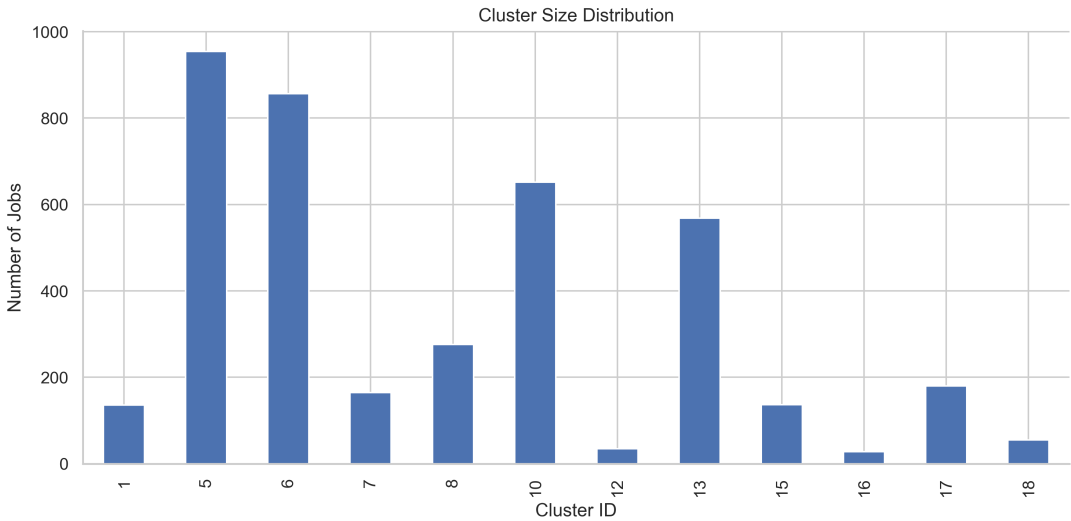
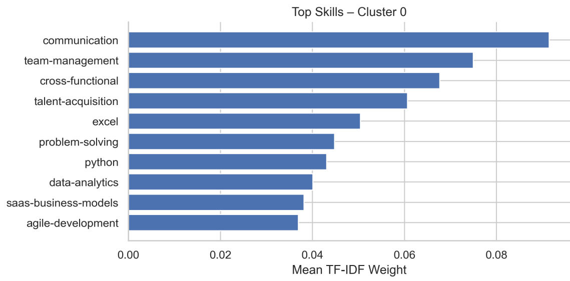


Figure 4.1: Number of jobs assigned to each of the 20 clusters. The uneven distribution reflects differing prevalence of skill profiles across the dataset.

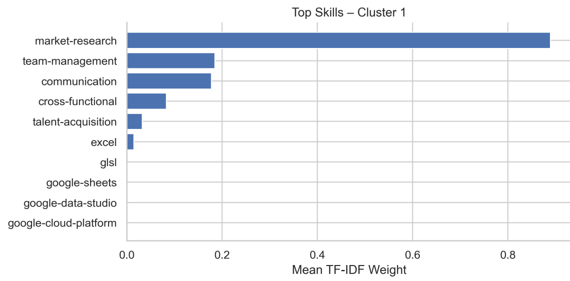
The difference is not surprising. Broad technological skill sets, such as generic software development and data analysis, are common across many businesses and industries, resulting in enormous clusters (4). Specialised roles, on the other hand, are naturally less common, particularly those associated with certain engines or production pipelines. The distribution thus represents true labour market structure rather than clustering artefacts (7).

4.2 Dominant Skills and Cluster Interpretation

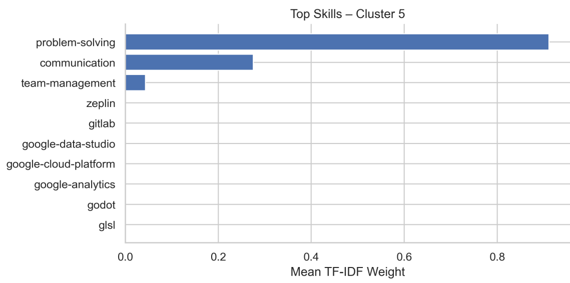
To assess interpretability, dominant skills within each cluster were identified by ranking mean TF-IDF weights (8; 9). Figure 4.2 presents representative examples from selected clusters.



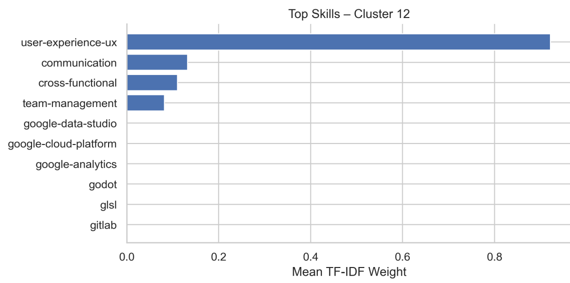
(a) Cluster 0



(b) Cluster 1



(c) Cluster 5



(d) Cluster 12

Figure 4.2: Top 10 dominant skills (by mean TF-IDF weight) for selected representative clusters, illustrating distinct skill profiles across gaming and technology roles.

The identified skill profiles align with intuitive role groupings (5; 6). Data-oriented clusters are characterised by skills such as *python*, *sql* and analytical libraries, while backend engineering clusters emphasise *java*, *spring boot* and web services. Infrastructure focused clusters are dominated by cloud and orchestration tools, including *aws*, *kubernetes* and *docker*.

Gaming-related clusters exhibit more distinctive patterns, with strong emphasis on engines and creative tools such as *unity*, *c#* and *blender*. These profiles reflect established production pipelines in the gaming industry, supporting the validity of skill-based grouping (7). At the same time, overlap between clusters is evident, particularly among general technology roles, reinforcing the prevalence of hybrid skill sets (4; 3).

4.3 Visualisation of Skill Space

Figure 4.3 presents a t-SNE projection of the reduced skill space, providing a qualitative view of cluster structure and overlap (17).

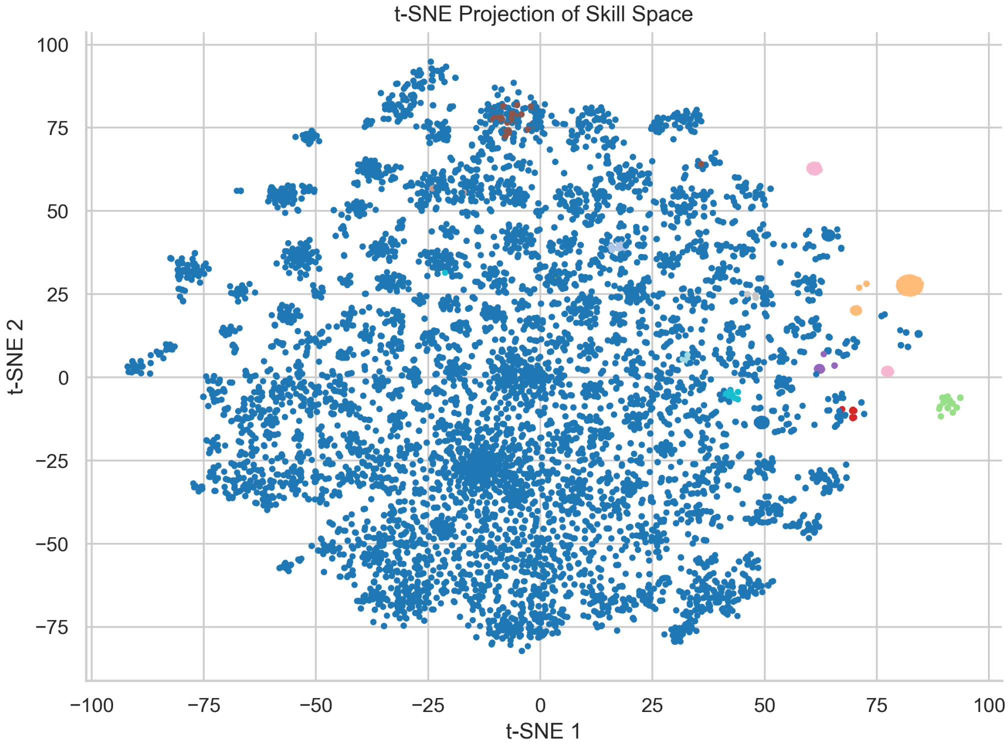


Figure 4.3: t-SNE projection of the reduced skill space coloured by K-Means cluster labels. Overlapping regions indicate hybrid and interdisciplinary roles.

The projection shows significant overlap between clusters, especially in regions dominated by broadly applicable abilities. Rather of forming sharply segregated groups, the data show progressive shifts, indicating common underlying competencies across roles. This structure explains the modest silhouette score and demonstrates the difficulties of rigid categorical boundaries in portraying modern employment positions (14; 3).

Gaming-related clusters look more compact in the projection, indicating a clearer separation caused by specialist toolchains. Non-gaming technology roles, on the other hand, show a greater degree of diversity, indicating greater adaptability and interdomain integration (4).

4.4 Confidence Score Distribution

Confidence scores derived from distance based soft assignment provide insight into assignment certainty. Figure 4.4 shows the distribution of confidence values across all job postings.

The distribution is substantially skewed towards lower values, with a mean confidence level of around 0.21. Only approximately 6% of jobs achieve the high confidence level of 0.5. Rather than demonstrating weak grouping, this result illustrates the intrinsic

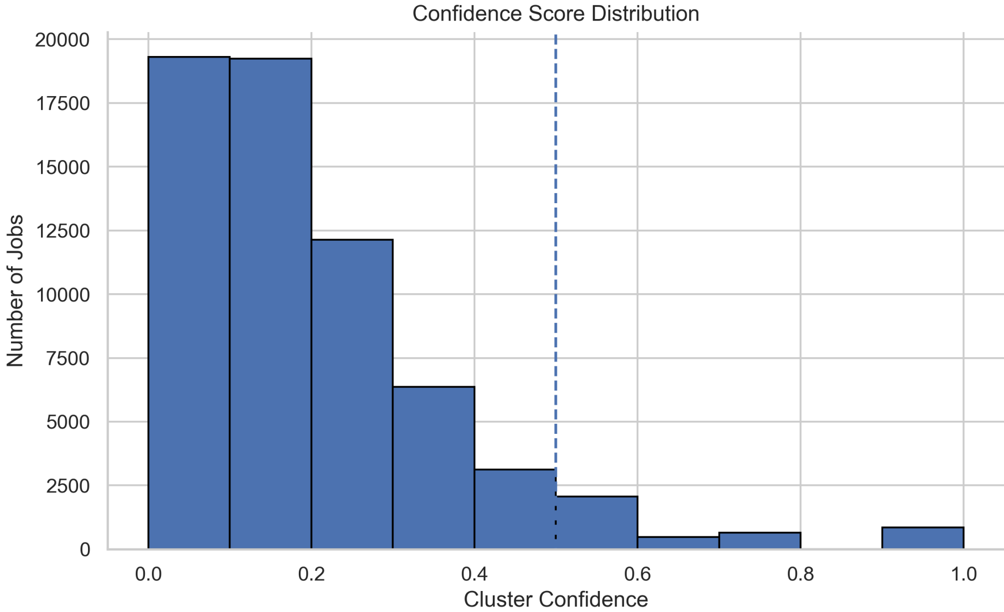


Figure 4.4: Distribution of confidence scores in 0.1 bins. The dashed line at 0.5 marks the threshold for high confidence assignments.

hybridity of many positions (3; 4). High confidence assignments relate to postings with specific and specialist skill requirements, whereas the majority of roles come from several clusters.

Explicitly modelling this uncertainty is a significant contribution of this work (18). By distinguishing between clear and ambiguous circumstances, the framework avoids imposing rigid labels on roles that do not fit into defined categories (18; 22).

4.5 Alignment with Existing Job Categories

Alignment between the derived clusters and original platform-defined job categories was evaluated using Kendall tau rank correlation (5). Figure 4.5 visualises the percentage overlap between original categories and clusters.

Moderate correlations suggests a partial agreement between the two representations. Where there is strong alignment, traditional categories capture relevant distinctions (1; 2). Where alignment is weak, the results show significant variation within categories, implying that broad labels mask large changes in skill composition (4; 7).

4.6 Implications for Stakeholders

Beyond numerical evaluation, the results of the skill-based clustering and confidence-aware taxonomy have practical consequences for a wide range of players in the gaming and technology ecosystem (7). The observed cluster patterns, confidence distributions, and partial agreement with existing categories shed light on how certain groups could benefit from more complex representations of employment functions.

For recruiters and hiring managers, the data emphasise the limitations of obtaining tal-

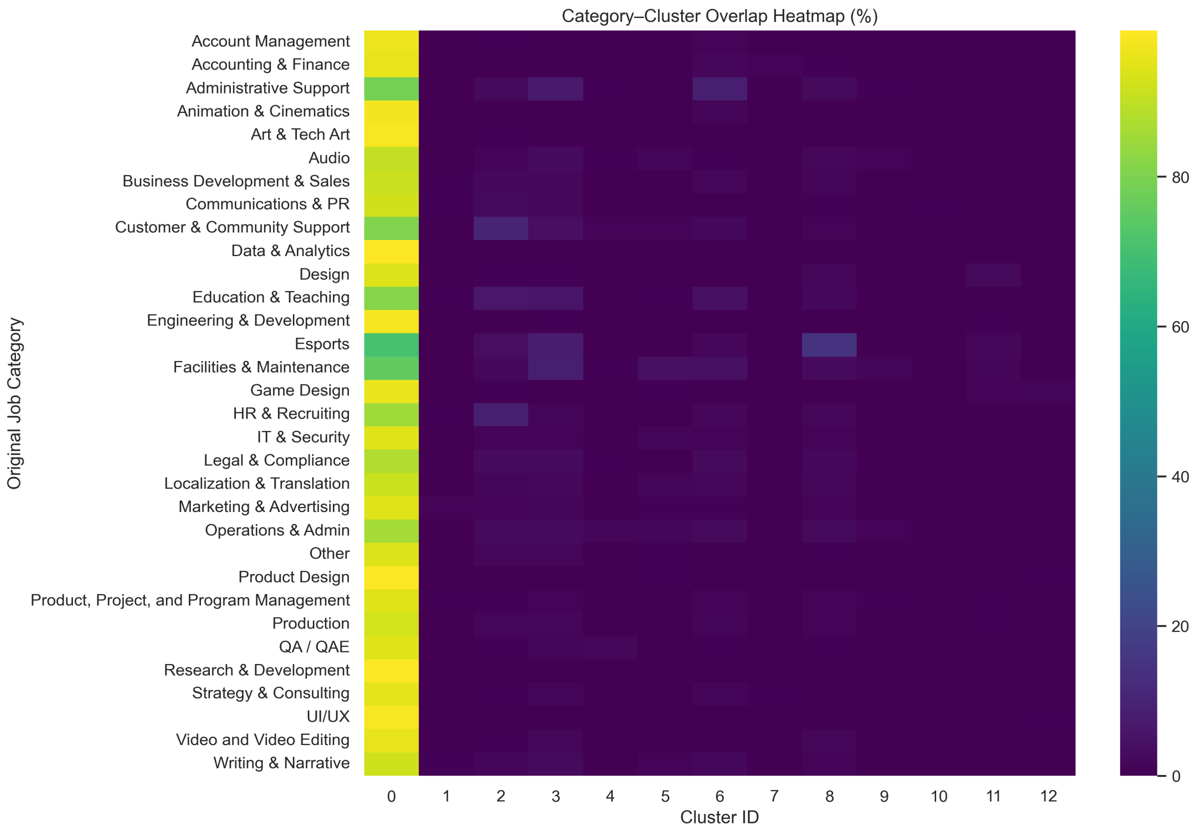


Figure 4.5: Overlap between original job categories and derived skill-based clusters. Rows sum to 100%, illustrating partial alignment and intra-category heterogeneity.

ent only through specific platform categories (1; 2). The vast, varied clusters and low fraction of high confidence assignments suggest that many positions require numerous skill domains (4). A confidence-aware taxonomy allows recruiters to find candidates who strongly correlate with specific skill profiles while also detecting hybrid positions that do not fit cleanly into traditional classifications. This allows for more flexible shortlisting and lowers the danger of eliminating qualified candidates owing to tight classification criteria (21).

According to the findings, skill-based representations can help with job discovery and self-positioning (7). Rather of navigating wide or unclear categories, candidates can have a better understanding of how their skill sets connect with dominating clusters and where their profiles may intersect. The precise modelling of uncertainty decreases the probability of candidates being misrepresented by a single label, instead recognising the breadth of their competencies (22).

For researchers and labour market experts, the findings highlight the importance of expanding beyond title- or category-based analysis (7). The intermediate correlations found between generated clusters and existing categories indicate both areas of agreement and significant internal heterogeneity (4). As a result, skill-based clustering offers a more comprehensive analytical lens for examining workforce structure, skill demand and job evolution in dynamic industries like gaming and technology (3; 4).

Finally, for platform designers and policymakers, the findings highlight the potential benefits of introducing confidence-based processes into job classification systems (22). Rather than pushing difficult tasks, platforms may highlight ambiguity as an instructive signal, promoting openness and more ethical use of automated categorisation in job contexts (22; 21).

4.7 Summary of Quantitative Results

Table 4.1 summarises key quantitative outcomes of the clustering and confidence analysis.

Table 4.1: Summary of key clustering and confidence results.

Metric	Value
Number of clusters	20
Overall silhouette score	0.0696
Mean confidence score	0.2059
Median confidence score	0.1821
High-confidence assignments (≥ 0.5)	6.29%

Overall, these findings show that skill-based clustering reveals interpretable structure beyond platform-defined categories, whereas confidence-aware assignment gives a more accurate portrayal of hybrid and multidisciplinary jobs (5; 18).

4.7.1 Linking Results to the Research Questions

The findings provide direct evidence for the three research questions posed in Chapter 1. First, the dominant-skill profiles within clusters indicate that skill-based grouping can yield coherent and interpretable job groupings beyond platform-defined categories, particularly where specialised toolchains are present (5). Second, the results illustrate that representation and modelling choices shape both separability and interpretability: TF–IDF supports inspection of clusters via skill weights (8; 9), while dimensionality reduction stabilises clustering in sparse, high-dimensional settings (16). Third, the confidence-aware mechanism adds practical value by distinguishing clear assignments from hybrid or ambiguous cases (18), with the observed confidence distribution suggesting that role hybridity is a common characteristic rather than an exception in these industries (3; 4).

4.8 Extended Evaluation and Robustness Discussion

Evaluating unsupervised clustering outcomes in labour market and skill-based analysis involves unique obstacles that differ from those found in supervised learning tasks (5). Unlike classification issues with ground-truth labels, clustering quality must be understood in terms of domain characteristics, representation choices, and analytical goals rather than absolute numerical criteria (19). The evaluation results in Chapter 4 reflect the structure of modern job data, rather than indicating methodological failure (3; 4).

The overall silhouette score obtained in this study is low when compared to values commonly reported in clustering tasks employing well-separated or falsely labelled datasets (14). However, this result is consistent with expectations for skill-based job advertisement data (7; 6). Job responsibilities in the gaming and technology industries usually demand overlapping and hybrid skills, resulting in subtle transitions across professional profiles rather than strongly demarcated categories (4). As a result, distance-based separation metrics, such as the silhouette score, naturally penalise such overlap, even when the resulting clusters are semantically relevant and understandable (14).

Importantly, the goal of this research is not to maximise numerical cluster separation, but to provide an interpretable taxonomy that depicts real-world patterns of skill demand (19). From this standpoint, the discovery of overlapping clusters and moderate separation provides empirical proof of the multidisciplinarity that characterises modern technology roles (3). Visual inspection of the reduced feature space confirms this interpretation, revealing continuous regions rather than discrete, compact clusters (17). These findings support the thesis that inflexible occupational boundaries are becoming increasingly out of sync with labour market reality (1; 2).

Robustness in unsupervised learning should be measured in terms of stability and interpretability, rather than predicted accuracy (19). This study demonstrates robustness through a variety of complementing aspects. First, dominating skill profiles within clusters remain consistent and intuitive despite exploratory changes in preprocessing thresholds and dimensionality reduction choices. Second, using TF–IDF representations ensures that cluster interpretations are based on observable skill tokens rather than latent features, which can be difficult to justify or audit (8; 9). Third, the confidence-aware assignment mechanism improves robustness by explicitly representing uncertainty, allowing ambiguous circumstances to be highlighted rather than implicitly absorbed into potentially misleading hard assignments (18; 22).

Clustering-based analysis is inherently sensitive to model selection. Decisions on cluster size, feature filtering thresholds, and dimensionality reduction parameters can all have an impact on cluster composition and assessment metrics. Rather than aiming to eliminate this vulnerability, the methodology used here prioritises transparency and reproducibility (19). Design decisions are explicitly documented, and their ramifications are explained in light of analytical objectives. This method is consistent with best practices in exploratory data analysis, where understanding the consequences of modelling assumptions is more important than presenting a single, optimised configuration (19).

The comparison of the resulting skill-based clusters to current platform-defined job categories strengthens the proposed taxonomy (1; 2). Moderate alignment suggests that the grouping captures identifiable occupational structure, whereas deviations emphasise variability within broad categories that traditional taxonomies fail to reveal (4). This balance of alignment and divergence indicates that the model is not arbitrary or too limited by pre-existing labels (1).

Overall, the evaluation findings show that the suggested framework accomplishes its goal of avoiding the false confidence associated with rigid classification while showing interpretable structure in complicated, overlapping skill data (18; 22). A more realistic and nuanced evaluation of job classification in dynamic industries is produced by combining quantitative indicators, qualitative examination, and confidence-aware interpretation (22).

Chapter 5

Critical Evaluation and Ethical Considerations

This chapter critically assesses the proposed skill-based, confidence-aware taxonomy and considers its ethical implications when applied to real-world labour market scenarios (18; 22). While the findings show that data-driven clustering can uncover substantial structure outside traditional job classifications, the strategy has methodological, data-related and practical limitations. Furthermore, the use of automated categorisation methods in recruiting and workforce analysis poses ethical problems about fairness, openness and responsible application (20; 21).

A fundamental shortcoming of the approach stems from the nature of job posting data. Skill data obtained from posts is inherently noisy and unreliable (7; 6). Employers employ different vocabulary to define comparable competences, including generic skills that have limited discriminatory power and may use specific to the platform templates with repeating artefacts. These characteristics contribute to sparsity and overlap in the feature space, limiting the degree of separation that can be achieved through clustering. The moderate silhouette score reported indicates this underlying structure rather than a failure of the clustering algorithm (14).

Decisions made during preprocessing have an even greater impact on outcomes. Tokenisation, filtering thresholds and artefact management decisions all entail balancing noise reduction and data accuracy (6). For example, deleting repeated or ubiquitous tokens may enhance numerical separation but risk ignoring signals that represent real-world posting habits. Retaining such items protects authenticity but may blur cluster boundaries. These trade offs emphasise the significance of clear documentation and sensitivity analysis when developing skill-based representations (19).

The choice of feature representation also imposes limits. TF-IDF vectorisation prioritises interpretability by preserving a direct mapping between characteristics and observable skills (8; 9). However, it lacks semantic awareness and regards lexically distinct but conceptually equivalent skills as separate. This constraint can fragment representations and dilute similarity signals, especially in fields with fast expanding vocabulary. While contextual embedding models may help to address these concerns, they also bring challenges in terms of explainability, reproducibility and computing expense (10; 11; 12). The chosen form shows a conscious preference for transparency over maximum semantic expressiveness (18; 19).

Clustering approach has other restrictions. K-Means presupposes nearly spherical clusters and equal variance, which are unlikely to hold true in varied skill spaces (13). Although dimensionality reduction mitigates the impacts of high dimensionality, irregular cluster geometry is still inaccurately represented. Density-based solutions, such as HDBSCAN, overcome some of these issues, but have proven unstable and difficult to interpret in this setting (15). As a result, the accepted strategy represents a practical compromise rather than a theoretically perfect answer.

The confidence-aware approach described in this paper addresses some of the constraints of hard clustering while also introducing interpretive considerations. Confidence ratings are calculated using distance-based probabilities and are affected by parameter choices such as temperature and threshold selection. These characteristics were empirically selected to balance sensitivity and usability, although they do not correlate to objective conceptions of correctness. Confidence should be viewed as relative alignment with cluster centroids rather than absolute categorisation certainty (18).

From an ethical standpoint, using automated taxonomies in recruitment and workforce analysis raises serious difficulties (20; 21). Systems that are trained on historical job data run the danger of repeating existing biases in labour markets (3). If specific abilities, occupations, or regions are under-represented due to systemic inequities, clustering may exacerbate their marginalisation by assigning them a low confidence or uncertain status. Without proper management, such systems may unintentionally disfavour candidates with non-standard or emergent skill sets (22).

The excessive use of automation poses an additional ethical issue. When algorithmic outputs are perceived as authoritative rather than advising, there is a risk of misuse in hiring, screening, and workforce planning (21). Low-confidence assignments can be misunderstood as signs of poor fit rather than hybridity or creativity. The explicit modelling of uncertainty in this paradigm helps to reduce this danger, but responsible use still necessitates clear communication of limitations and the maintenance of human judgement (18).

The connection between interpretability and fairness is likewise difficult. While interpretable models promote transparency, they do not necessarily prevent biassed outcomes (19). Dominant skill patterns may represent industry conventions that promote specific educational backgrounds, career pathways, or geographies. Ethical deployment necessitates not only clear technique, but also critical evaluation on whose abilities are represented and whose may be disregarded (20).

Legal and regulatory factors emphasise the need for care. In jurisdictions with data protection and algorithmic accountability frameworks, systems that influence employment choices may be required to be explainable and contestable (22). Although the suggested taxonomy is designed as an analytical and support for decisions tool rather than an automated decision system, its potential downstream applications demand careful framing and documentation (21).

In conclusion, the proposed taxonomy should be viewed as an exploratory and supporting analytical framework rather than a final classification. Its strength is in improving knowledge of skill hierarchies and emphasising uncertainty rather than imposing rigid classification. Ethical application requires preserving transparency, identifying limitations and ensuring that automated insights supplement rather than replace human expertise (18; 22).

5.1 Responsible Use and Governance Implications

While the suggested skill-based, confidence-aware taxonomy is meant as an analytical and support for decisions tool, its potential use in recruiting and workforce analysis raises important questions about responsible use and governance (21). Automated categorisation systems, particularly those used in job settings, pose inherent hazards when outputs are perceived as authoritative rather than instructive (20).

A key problem is automation bias, in which human decision-makers place unwarranted reliance in computational outcomes (22). In the context of job classification, low-confidence or unclear assignments may be misunderstood as markers of poor candidate appropriateness rather than reflections of interdisciplinary talent profiles. This framework's explicit modelling of uncertainty contributes to mitigating this risk by highlighting ambiguity as an informative signal (18). However, effective deployment demands that confidence scores be properly disclosed and perceived as relative alignment metrics rather than final classifications.

Governance measures are thus required when integrating such technologies into existing platforms. These could include humans in the loop methods, transparent description of model limits, and safeguards against automated exclusion based on strict thresholds (21). The suggested framework is designed to promote these goals by promoting interpretability and avoiding black-box representations that obscure decision logic (19).

5.2 Interpretability, Fairness, and Bias Considerations

Although interpretability is an important advantage of the suggested approach, it does not guarantee fairness (19). Skill-based grouping reflects trends in the underlying data, which may contain structural biases within labour markets (3). If some roles, geographies or demographic groups are under-represented in job advertisements, the resulting taxonomy may be disproportionately reflective of dominant sector norms (7).

In the gaming and technology sectors, this risk is particularly salient due to uneven access to formal credentials, tooling, and geographic opportunities. Clusters characterised by specialised or emerging skills may appear less cohesive or receive lower confidence scores, not because they lack relevance, but because they are less frequently represented in the data. Without careful interpretation, such outcomes could reinforce existing inequalities by marginalising non-standard or innovative skill combinations (20).

The usage of TF-IDF representations promotes transparency by allowing dominant skills to be evaluated directly (8; 9). This allows for careful analysis of cluster composition and bias audits at the feature level. Nonetheless, ethical application of the taxonomy necessitates a constant assessment of whose skills are magnified and which may be neglected (22). As a result, the framework should be seen as a beginning point for further investigation rather than a definitive depiction of labour market worth.

5.2.1 Bias Amplification and Mitigation Strategies

While skill-based clustering provides a data-driven alternative to traditional job categorisation, it inherits biases from the underlying job advertisement data (7). Online job advertisements reflect historical, organisational and cultural trends, such as unequal access to opportunities, regional concentration of industries, and differences in how companies characterise identical employment (3). As a result, clustering models trained on such data run the danger of exacerbating existing discrepancies if their results are interpreted without proper contextual knowledge (20).

One cause of bias is the uneven frequency with which skills are mentioned. Skills that are highly sought or commonly mentioned by employers may dominate feature representations, whereas new or less standardised abilities may be poorly represented (6). This imbalance can lead to clusters favouring conventional career paths, thereby marginalising non-traditional or multidisciplinary profiles. Furthermore, specific to the platform conventions and templates can introduce artefacts that influence similarity metrics in ways unrelated to true skill requirements (6).

Geographic and sectoral prejudice should also be considered. Job postings are frequently concentrated in specific locations or industries, resulting in excessive representation of certain labour markets (7). In the game and technology industries, this may favour jobs associated with major studios or international corporations while underestimating smaller companies or individual developers. As a result, the generated taxonomy should not be taken as a universal or comprehensive description of all employment roles within these areas (3).

Several design choices in the proposed framework are intended to alleviate, rather than eliminate, these hazards. The use of interpretable TF-IDF representations enables direct inspection of dominant skills inside clusters, allowing for feature-level bias auditing (8; 9). This openness allows us to identify scenarios in which clusters are driven by generic or artifactual terms rather than genuine competencies. Furthermore, thorough documenting of preprocessing decisions facilitates reproducibility and critical examination of model assumptions (19).

The confidence-aware assignment approach protects against bias amplification. By recognising low-confidence assignments, the approach avoids putting diverse or atypical roles into dominant clusters, where they may be distorted. Instead, ambiguity is presented as an informative signal, promoting human interpretation over automatic exclusion (18; 22). This is especially significant in recruitment contexts, where rigid classification can disfavour candidates with skill profiles that do not fit traditional models (21).

Regardless of these mitigation techniques, it is crucial to recognise that no purely data-driven methodology can completely eliminate bias from labour market analysis (20). Responsible use necessitates a combination of artificial insights, domain expertise and human monitoring (22). The proposed taxonomy should be seen as a complimentary analytical tool that facilitates exploration and comprehension, rather than a final or prescriptive decision-making system.

5.3 Limitations of Ethical Safeguards

While the trusted system explicitly recognises uncertainty, it does not eliminate all ethical risk. Distance-based metrics are used to calculate confidence ratings, which are further influenced by modelling assumptions, parameter selection and feature representation (18). As a result, they represent relative similarity within the built environment rather than objective criteria of fitness or competency.

Furthermore, the taxonomy does not take into consideration contextual elements such as experience level, career growth or informal skill acquisition, all of which are often important in hiring decisions (4). Ethical deployment hence involves complementing assessment techniques and subject knowledge. The framework is better suited for exploration, comparison and insight development rather than automated screening or ranking (22).

In conclusion, the ethical application of the suggested taxonomy is dependent not just on technical design decisions, but also on how outputs are presented, interpreted and integrated into larger systems for making decisions (21). When applied responsibly, the technique provides a transparent and adaptable alternative to strict classification schemes, while also recognising the complexity and diversity of current job functions (18; 22).

Chapter 6

Conclusion and Future Work

This dissertation studied whether job classification in the gaming and technology industries may be improved by shifting away from strict, platform-related labels and towards a skill-based, data-driven taxonomy that explicitly allows for uncertainty (3; 1). Motivated by the growing frequency of hybrid and transdisciplinary professions, the study suggested and tested an interpretable clustering framework meant to better accurately reflect modern skill need (4).

Addressing the first study question, the findings show that skill-based clustering can reveal coherent and meaningful groupings that correspond to underlying toolchains and competency profiles (5; 6). Compared to typical platform-defined categories, generated clusters reveal greater distinctions and significant variation within existing labels. This is especially true in industries like gaming, where skill combinations that are not covered by general occupational categories are constrained by production pipelines (7).

Regarding the second research question, the findings emphasise how crucial methodological decisions are in determining the final taxonomy's quality and usability. While dimensionality reduction using Truncated SVD lessened the difficulties of high dimensionality (16), the adoption of TF-IDF representation prioritised interpretability, enabling clusters to be directly explained using dominating skills (8; 9). Despite having moderate internal evaluation scores that indicate significant talent overlap rather than methodological failure, K-Means clustering offered a reliable and understandable grouping mechanism (13; 14). These results highlight the need of interpreting clustering measurements contextually, especially in labour market sectors where position transitions are gradual (3).

The role of uncertainty in job classification was the subject of the third study question. The main contribution of this dissertation is the establishment of a reliable assignment mechanism. The framework differentiates between roles that hold ambiguous or hybrid positions and those that closely align with a dominating skill group by converting distances to cluster centroids into probabilistic confidence scores (18). The finding that just a tiny percentage of jobs achieve a high confidence criteria is not a flaw; rather, it provides insight into the multidisciplinary character of contemporary labour (4). By explicitly quantifying this ambiguity, more nuanced interpretation of clustering outcomes is supported and the misleading accuracy inherent in hard classification is avoided (22).

In addition to answering the research goals, this study offers a transparent and repeatable methodological pipeline for large scale skill data analysis. The approach is appropriate for exploratory labour market analysis and decision support due to its emphasis on interpretability, documented design decisions and uncertainty estimation (19). Enhancing job recommendation systems, assisting with workforce analytics, and influencing curriculum design by spotting new talent clusters and regions of overlap are some possible uses (7).

The highlighted constraints provide several directions for future development. Incorporating contextual embedding models could increase semantic coherence and reduce fragmentation caused by lexical variance, but care must be taken to maintain explainability (10; 12). Enriching representations with extra attributes such as job titles, experience levels or income information may increase clustering quality, depending on data availability and consistency. Particular domains or hierarchical clustering techniques could also be investigated to better capture global structure and sectoral variation.

External validation is an essential option for expansion. Cluster coherence evaluation by industry practitioners would bolster claims of practical relevance, whereas application to geographically diverse or multilingual datasets would improve generalisability. Finally, more study into approaches for bias reduction and responsible deployment is required if such taxonomies are to be employed in real-world recruitment or workforce planning situations (20; 21).

In conclusion, this dissertation illustrates that data-driven, confidence-aware clustering is a credible and adaptable alternative to inflexible job categorisation in rapidly changing sectors. By accepting uncertainty and emphasising interpretability, the proposed framework provides a more accurate portrayal of modern job positions, laying the groundwork for more responsive and responsible labour market analysis (18; 22).

6.1 Practical Implications and Responsible Use

The proposed taxonomy is best interpreted as a decision-support and analytical framework rather than an automated decision system (21). For recruitment and hiring workflows, it can support more transparent exploration of role similarities by highlighting dominant skill patterns, while the confidence scores can be used to flag cases where a single label is likely to be misleading due to hybridity. For job platforms and labour market analysts, the framework can complement existing categories by surfacing intra-category heterogeneity and cross-category overlap that static taxonomies may obscure (1; 2).

However, responsible use requires careful communication of limitations. In particular, confidence scores should not be interpreted as a measure of correctness, candidate suitability, or job quality; they represent relative alignment with cluster centroids under the chosen representation and modelling assumptions (18). Where used in practice, the framework should be accompanied by human oversight, transparency about data and parameter choices, and safeguards to avoid automation bias or unfair exclusion based on rigid thresholds (22; 21).

Chapter 7

Discussion

This chapter situates the dissertation's findings within the broader literature on job classification, skill-based analysis, and labour market modelling (3; 7). Rather than restating the findings, the discussion focusses on assessing their importance, explaining differences from previous research, and reflecting on what the observed patterns say about current job structures in the gaming and technology industries.

A key conclusion of this study is the degree of skill overlap across job roles which is reflected in moderate clustering separation and a low number of high-confidence assignments (4). This result differs from typical job taxonomies, which implicitly presume that occupational boundaries are well defined and stable (1; 2). Instead, the findings lend credibility to previous claims that modern technology occupations are fundamentally hybrid, incorporating competencies from many functional domains (3). In this regard, the observed ambiguity should not be interpreted as a methodological flaw, but rather as evidence that inflexible categorisation frameworks are becoming increasingly out of sync with labour market reality (3; 4).

Compared to previous skill-based clustering research, which frequently show higher internal evaluation metrics (5; 6), the current findings emphasise the importance of representation and interpretability limitations. Studies that use deep semantic embeddings usually enhance numerical separation, albeit at the expense of transparency (10; 11; 12). In contrast, the usage of TF-IDF in this dissertation prioritises explainability, allowing clusters to be directly interpreted using dominant skills (8; 9). The trade off shown here supports the thesis that performance measurements alone are insufficient for evaluating job taxonomies designed for practical usage (19).

The confidence-aware assignment mechanism stands out from other approaches. Most previous research requires strict cluster membership, implicitly interpreting uncertainty as noise (5). In contrast, this study views ambiguity as an important signal (18). The discovery that only a tiny percentage of occupations are firmly aligned with a single cluster highlights the ubiquity of multidisciplinary job design, particularly in non-gaming technology industries (4). This is consistent with broader labour market trends towards role flexibility and continual skill recombination (3).

The differences identified between gaming and non-gaming roles deserve special consideration. Gaming-related clusters showed stronger coherence and separation, most likely due to the structured nature of game production pipelines and reliance on specialised engines and tools (7). Non-gaming technology roles, on the other hand, have a high degree of overlap, indicating greater flexibility in how talents are integrated across businesses (4). This divergence shows that domain-specific taxonomies may have advantages over universal models, which has gotten little attention in previous research (1).

The alignment study highlights the limits of defined by platforms categories. A moderate correlation between computed clusters and existing labels suggests some agreement, but also demonstrates significant variety within categories (4). This study backs up criticisms of broad occupational categories and indicates how skill-based approaches can supplement rather than replace traditional taxonomies by highlighting internal heterogeneity and crossover similarities (3; 1).

Unexpected outcomes also merit discussion. The low silhouette score, while initially surprising, was consistent with a visual evaluation of the skill area (17). Rather than weakening the clustering strategy, this result emphasises the constraints of using internal metrics for evaluation in high-dimensional, overlapping domains (14). The combination of quantitative measures, visualisation and interpretability analysis results in a more balanced assessment of clustering quality (19).

From a practical standpoint, the findings suggest that job classification systems should stop treating categories as mutually exclusive. Instead, flexible representations that acknowledge partial membership and uncertainty may be more useful for recruitment, recommendation and workforce analytics (21).

7.1 Domain-Specific Patterns and Implications for Taxonomy Design

The investigation revealed an important distinction between gaming-related and non-gaming technology tasks in terms of cluster coherence and separation. Gaming-related clusters have a higher level of internal consistency which can be attributed to the structured nature of game production pipelines and the use of specialised engines and tools (7). Technologies like game engines, graphics frameworks and asset development software enforce relatively fixed skill combinations, resulting in more distinct and consistent skill profiles.

On the other hand, there is far more overlap between clusters in non-gaming technological roles. Because these competencies are modular and transportable, skills related to software development, data analysis, cloud infrastructure and system design regularly co-occur across a wide range of employment areas (4). The flexible skill composition of modern technology work is more effectively represented by this flexibility which also helps to reduce cluster fragmentation. Therefore, rather than being a drawback of the clustering approach, the observed overlap should be understood as a characteristic of the employment market (3).

These specific to the domain patterns have ramifications for the creation and use of job taxonomies. Universal classification schemes that use the same basic assumptions across industries risk hiding important differences in how jobs are arranged and described (1; 2). The findings indicate that sector-aware or specific taxonomies may better capture the realities of skill demand, particularly in businesses with rapid technological change (4).

More broadly, the findings support the notion that occupational categorisation should shift away from strict, mutually exclusive designations. Flexible representations that account for partial membership, uncertainty and skill recombination are more suited to dynamic labour markets (5). By explicitly describing these qualities, the proposed framework lays the groundwork for more responsive and realistic approaches to workforce analysis and job classification (18).

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Appendix A

Appendix

A.1 Tools and Computational Environment

All experiments and analyses in this dissertation were implemented using Python 3.12. The following libraries and tools were used throughout the project:

- **Data handling and preprocessing:** `pandas`, `numpy`
- **Feature extraction and clustering:** `scikit-learn` (TF–IDF Vectorizer, K-Means, Truncated SVD, t-SNE, silhouette score)
- **Density-based clustering (exploratory):** `hdbscan`
- **Visualisation:** `matplotlib`, `seaborn`

All experiments were conducted on a standard laptop environment without GPU acceleration. To ensure reproducibility, a fixed random seed (42) was used consistently across preprocessing, dimensionality reduction, and clustering stages. The complete implementation is provided in the accompanying Jupyter notebook `Final Production.ipynb`.

A.1.1 Dataset Versions and Preprocessing Summary

Table A.1 summarises the dataset size before and after preprocessing and validation.

Table A.1: Summary of dataset versions before and after preprocessing and validation.

Dataset version	Number of postings	Notes
Raw dataset	70,017	Collected job advertisements with platform metadata and skill strings.
Filtered/cleaned dataset	64,230	Removed incomplete entries and applied preprocessing steps described in Chapter 3.

A.2 Representative Code Excerpts

This section presents selected code excerpts to illustrate the core components of the proposed pipeline. These snippets are representative rather than exhaustive; the full implementation is available in the project notebook.

A.2.1 Skill Preprocessing and Vectorisation

```
df['Skills'] = (
    df['Skills']
    .fillna('')
    .str.lower()
    .str.replace(r'\s+', ' ', regex=True)
)

vectorizer = TfidfVectorizer(
    token_pattern=r'(?u)\b\w[\w-]+\b',
    min_df=5,
    max_df=0.95
)

X = vectorizer.fit_transform(df['Skills'])
```

A.2.2 Dimensionality Reduction and Clustering

```
svd = TruncatedSVD(n_components=300, random_state=42)
X_reduced = svd.fit_transform(X)

kmeans = KMeans(
    n_clusters=20,
    init='k-means++',
    n_init=10,
    random_state=42
)

labels = kmeans.fit_predict(X_reduced)
```

A.2.3 Confidence Score Computation

```
distances = pairwise_distances(
    X_reduced,
    kmeans.cluster_centers_
)

probs = np.exp(-distances)
probs /= probs.sum(axis=1, keepdims=True)

confidence_scores = probs.max(axis=1)
```

A.3 Sample Outputs

A.3.1 Dominant Skills per Cluster

Table A.2 presents dominant skills for selected clusters. The full cluster summaries are available in the exported file `FINAL_cluster_summary_v2.csv`.

Table A.2: Example dominant skills for selected clusters (top five by mean TF–IDF weight).

Cluster	Dominant Skills
0	python, sql, data-analytics, pandas, numpy
1	unity, c-sharp, 3d-modeling, blender, game-design
5	java, spring-boot, restful-web-services, maven
12	aws, kubernetes, docker, ci-cd, terraform

A.3.2 Sample Job Assignments

Table A.3 provides example job postings with assigned clusters and confidence scores.

Table A.3: Sample job assignments with cluster labels and confidence scores.

Job Title	Category	Cluster	Confidence
Senior Java Consultant	Programming	5	0.61
Tech Lead (Angular)	Frontend Development	3	0.54
3D Game Artist	3D Art	1	0.72
Data Engineer	Data Analysis	0	0.48
Senior Product Designer	Product Design	8	0.39

Appendix B

Code Availability

The following repository provides public access to the whole implementation used in this dissertation, including evaluation notebooks, feature extraction, clustering pipelines, and data pretreatment scripts:

`https://liveastonac-my.sharepoint.com/:f:/r/personal/240401328_aston_ac_uk/Documents/Dissertation_Final?csf=1&web=1&e=HT8RFX`

All of the code needed to replicate the tests and findings presented in this dissertation is included in the repository.

B.1 Supplementary Visual Analysis

The analyses in the dissertation's main body are supported and contextualised by the additional visualisations in this part. By showing how platform-defined job categories link to the generated skill-based taxonomy and how domain-specific patterns emerge from the clustering findings, the pictures are presented to improve transparency and interpretability. These visualisations do not present any new experimental findings beyond those covered in previous chapters; rather, they are meant for further investigation and validation.

B.2 Final Model Configuration

The final configuration of the proposed taxonomy model is summarised below:

- Feature representation: TF–IDF
- Dimensionality reduction: Truncated SVD (300 components)
- Clustering algorithm: K-Means ($k = 20$)
- Overall silhouette score: 0.0696
- Mean confidence score: 0.2059
- High-confidence assignments (≥ 0.5): 6.29%

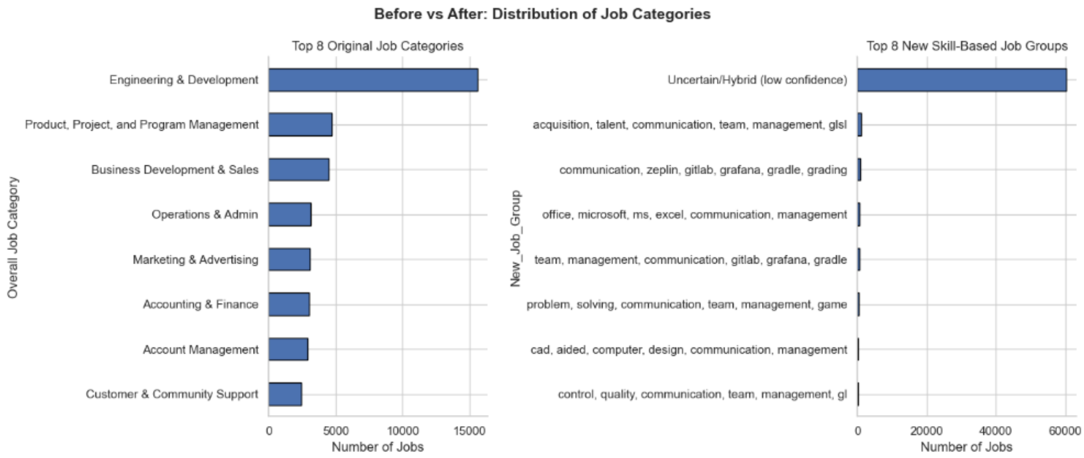

Figure B.1: Comparison between platform-defined job categories and derived skill-based job groups. The figure illustrates how a small number of broad platform categories are redistributed into more granular and heterogeneous skill-based groupings. This visualisation supports the motivation for adopting a skill-based taxonomy.

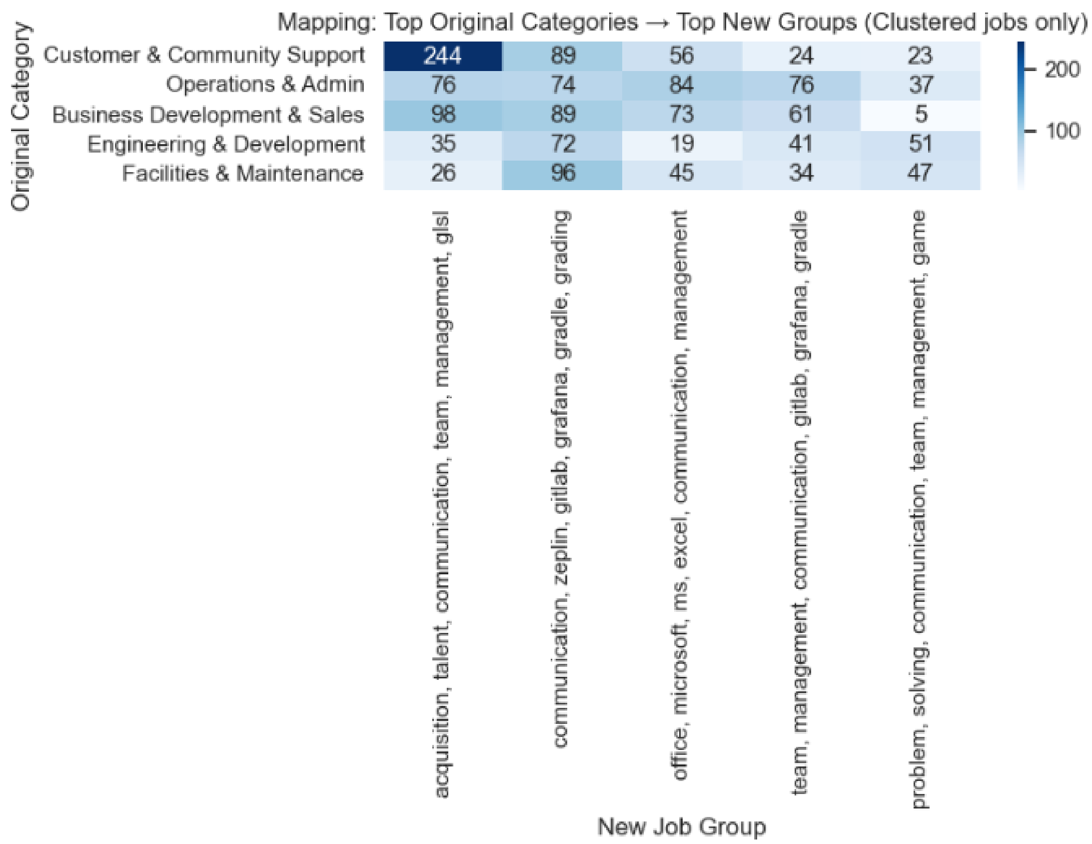

Figure B.2: Heatmap showing the mapping between selected platform-defined job categories and the most frequent derived skill-based job groups. The dispersion across multiple groups highlights the internal heterogeneity of traditional categories and the limitations of one-to-one category assignment.

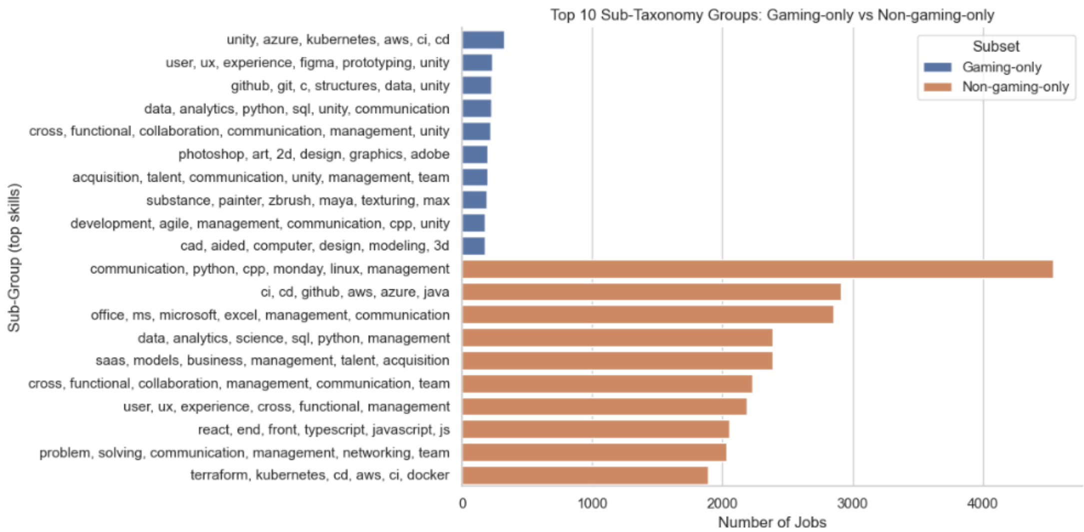

Figure B.3: Comparison of the most frequent sub-taxonomy groups derived from gaming-only and non-gaming-only job postings. Gaming-related roles exhibit more specialised and tool-centric skill combinations, while non-gaming roles show broader and more transferable skill profiles. This figure is included for supplementary domain-specific analysis.